HOSTEL & MESS MANAGEMENT SYSTEM

HOSTEL CONNECT | DBSE & DBD | TEAM ID: 22 | SECTION: S8

Pranav Chowdary – 2520030555 | R. Varshith Reddy – 2520030250
Dhanush Chandra – 2520030014

Instructor & Guide: Dr. D. Ramya Krishna | Course Coordinator: Dr. R. Sateesh Kumara

DATABASE SYSTEMS / FULL-STACK IMPLEMENTATION

Abstract

Hostel Connect is a database-driven full-stack platform for student accommodation and mess administration. It integrates room and bed allocation, occupancy, mess and meal attendance, fee and payment records, complaints, announcements, and role-based dashboards. The system uses React/Vite for the interface, Node.js/Express for REST APIs, and MySQL for persistent relational data. It centralizes hostel operations, reduces duplicate manual records, and improves access to structured information for students, wardens, and administrators. The platform also supports reporting, maintenance grievance tracking, and scalable database-backed management workflows.

Tools & Technologies

- Front-End: React + Vite, HTML/CSS, JavaScript
- Back-End: Node.js + Express.js
- Database: MySQL; SQL scripts / MySQL Workbench
- IDE: Visual Studio Code
- Architecture: REST API + relational database

Problem Statement & Objectives

Manual hostel records can become fragmented across rooms, fees, attendance and complaints. Hostel Connect centralizes these operations and maintains relationships through MySQL.

Objectives: centralized records • capacity-controlled allocation • mess/menu tracking • fee/payment management • complaint workflow • SQL reporting.

Existing vs Proposed + Feasibility

EXISTING

Manual records • scattered spreadsheets • duplicate data • limited reporting

PROPOSED

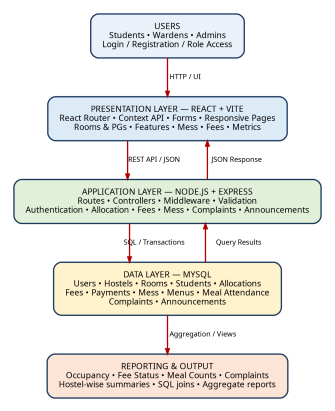
Centralized MySQL • role dashboards • automated allocation • reports

FEASIBILITY Technical • Operational • Economic • Schedule-ready

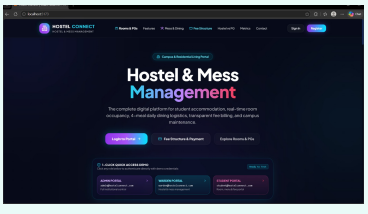
Key Technical Flow

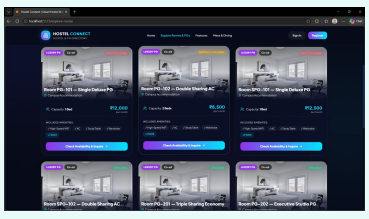
1. **Users / Roles**
 Admin • Warden • Student • login / registration
2. **Frontend**
 React + Vite • Router • Context API • responsive pages
3. **REST API**
 Express routes • controllers • validation • role middleware
4. **Database**
 MySQL • PK/FK constraints • joins • transactions
5. **Outputs**
 Occupancy • fees • meal counts • complaints • announcements

System Architecture

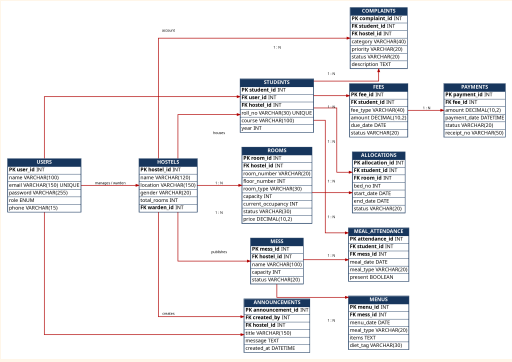


Implementation Screenshots





ER Diagram



Results & Testing

Centralized records reduce duplication. SQL joins generate organized reports. Testing covers authentication, allocation, fees, complaints and database persistence.

Conclusion

Hostel Connect connects the user interface, REST API and MySQL database into a single workflow for accommodation, mess, fees and resident support.

Future Scope

Online room booking • online fee payment • digital receipts • mobile access • notifications • QR/biometric meal attendance • analytics for occupancy and mess usage.